

Manit Garg

🌐 manitgarg.vercel.app | 📞 814-826-8819 | ✉️ manit.garg.in@gmail.com | in.linkedin.com/in/manitgarg | [Github.com/SinfulSoul007](https://github.com/SinfulSoul007)

EDUCATION

Pennsylvania State University

Aug. 2023 – May 2027

University Park, PA

- Bachelor of Science in Computer Science
- Business Fundamentals Certificate, Smeal College
- Computer Engineering, Minor
- Cybersecurity Computational Foundations, Minor

LEADERSHIP & INVOLVEMENT

Research Assistant, Explainable AI for Program Semantic Equivalence

Mar. 2025 – Present

Pennsylvania State University

University Park, PA

- Conduct research on AI-driven program semantic equivalence, focusing on explainability in detecting obfuscated and regular program similarities.
- Develop models utilizing graph-based learning techniques, such as Graph Convolutional Networks (GCN) and Graph Attention Networks (GAT).
- Implement and evaluate embedding techniques, including Word2Vec, Code2Vec, and DeepWalk, to analyze program structure.
- Assess model effectiveness using semantic equivalence accuracy, explainability scores, and obfuscation robustness.
- Contribute to a framework for enhancing program verification, plagiarism detection, and automated code review systems.
- Presented research findings at the university's Computer Science Research Symposium and contributed to a research paper currently under review.

Treasurer & Event Coordinator, Machine Learning @ Penn State (ML@PSU)

Sept. 2024 – Present

Pennsylvania State University

University Park, PA

- Manage ML@PSU's financial transactions, budget allocations, and fundraising efforts, ensuring compliance with university policies.
- Prepare financial reports and oversee funding applications to support organizational growth.
- Plan and coordinate events, workshops, and discussions on ML concepts and industry trends.
- Collaborate with university staff and stakeholders to facilitate impactful events and networking opportunities.

PROJECTS

Game Captch | Next.js, React, TensorFlow, WebXR

Apr. 2025

- * Developed an innovative CAPTCHA alternative at HackPSU Spring 2025 that verifies human users through an interactive space shooter game.
- * Implemented bot resistance using a pre-trained Deep Q-Network (DQN) model to benchmark and calibrate game difficulty.
- * Built with Next.js and Phaser.js, featuring VR compatibility for MetaQuest headsets via WebXR.
- * Created a modular, embeddable verification system with backend support through Railway and Supabase.
- * Won the 10th Anniversary: Timeless Tech Track at HackPSU Spring 2025.

Swipe: AI-Driven Professional Networking App | React, Node.js, MongoDB, AI APIs

Oct. 2024

- * Built a networking app that connects professionals based on shared research and career interests using AI-driven matching.
- * Designed a swipe-based interface for quick browsing of potential connections.
- * Added real-time messaging for instant, seamless networking and collaboration.
- * Created a user-friendly experience to bridge networking gaps in specialized fields.

Eye-Ballers: Glaucoma Detection Model | Python, Flask, React, StyleGAN3

Oct. 2023

- * Developed for HackPSU Fall 2023, Eye-Ballers is a glaucoma detection model leveraging StyleGAN3 to identify glaucoma in eye images. Combines a user-friendly frontend with a robust backend to provide real-time results.
- * **Frontend:** Intuitive interface for image upload and analysis.
- * **Backend:** StyleGAN3-based model processes images for glaucoma detection.
- * Gained valuable experience in healthcare innovation, teamwork, and overcoming technical challenges as a freshman team.

TECHNICAL SKILLS

Languages: Java, Python, C/C++, JavaScript, TypeScript, HTML/CSS, SQL

Frameworks: React, Node.js, Flask, WordPress, Material-UI, FastAPI, Express.js, Next.js

Developer Tools: Git, Docker, TravisCI, Google Cloud Platform, VS Code, PyCharm, IntelliJ, Jira

Libraries & Technologies: pandas, NumPy, Matplotlib, TensorFlow, PyTorch, scikit-learn, Jupyter, Redux